

# 3I/ATLAS Production Rate Table

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**Table 1:** Catalog of production rates for chemical species  $Q_X$  from observations of 3I/ATLAS. Includes error associated with  $Q_X$ , observation dates, heliocentric distance  $r_H$ , facility used to obtain observation, and relevant citation. Upper limit production rates are indicated with  $<$  and repeated entries are indicated with three dots. Note that the errors are of the same order of magnitude as the corresponding value of  $Q_X$ .

| Volatile         | Production Rate<br>[s <sup>-1</sup> ] | Error<br>[s <sup>-1</sup> ] | Observation Date  | $r_H$<br>[au] | Facility     | Reference                                     |
|------------------|---------------------------------------|-----------------------------|-------------------|---------------|--------------|-----------------------------------------------|
| H <sub>2</sub> O | $< 9.10 \times 10^{26}$               | 0.00                        | 07/04/25–05/26/25 | 4.37          | X-SHOOTER    | <a href="#">Alvarez-Candal et al. (2025)</a>  |
|                  | $7.40 \times 10^{26}$                 | 5.00                        | 07/31/25          | 3.51          | UVOT         | <a href="#">Xing et al. (2025)</a>            |
|                  | $3.20 \times 10^{26}$                 | 0.20                        | 08/01/25–08/15/25 | 3.20          | SPHEREx      | <a href="#">Lisse et al. (2025a)</a>          |
|                  | $2.23 \times 10^{26}$                 | 0.08                        | 08/06/25          | 3.32          | JWST/NIRSpec | <a href="#">Cordiner et al. (2025)</a>        |
|                  | $< 1.50 \times 10^{26}$               | 0.00                        | 08/08/25–08/12/25 | 3.10–3.30     | SPHEREx      | <a href="#">Lisse et al. (2025b)</a>          |
|                  | $1.36 \times 10^{27}$                 | 0.50                        | 08/19/25          | 2.90          | UVOT         | <a href="#">Xing et al. (2025)</a>            |
|                  | $< 9.46 \times 10^{27}$               | 0.00                        | 08/26/25–09/03/25 | 2.54          | TMRT         | <a href="#">Li et al. (2026)</a>              |
|                  | $1.45 \times 10^{28}$                 | 0.52                        | 09/08/25–09/09/25 | 2.27          | ...          | ...                                           |
|                  | $2.08 \times 10^{28}$                 | 0.41                        | 09/18/25–09/23/25 | 1.96          | ...          | ...                                           |
|                  | $8.00 \times 10^{28}$                 | 0.80                        | 11/02/25          | 1.36          | MAJIS/JUICE  | <a href="#">Bockelee-Morvan et al. (2026)</a> |
|                  | $< 1.60 \times 10^{29}$               | 0.00                        | 11/04/25          | 1.37          | ALMA/ACA     | <a href="#">Salazar Manzano et al. (2026)</a> |
|                  | $3.17 \times 10^{29}$                 | 0.004                       | 11/06/25          | 1.40          | SOHO/SWAN    | <a href="#">Combi et al. (2025)</a>           |
|                  | $2.31 \times 10^{29}$                 | 0.003                       | 11/09/25          | 1.42          | ...          | ...                                           |
|                  | $2.35 \times 10^{29}$                 | 0.004                       | 11/10/25          | 1.43          | ...          | ...                                           |
|                  | $2.36 \times 10^{29}$                 | 0.004                       | 11/12/25          | 1.45          | ...          | ...                                           |
|                  | $6.80 \times 10^{28}$                 | 0.70                        | ...               | 1.45          | MAJIS/JUICE  | <a href="#">Bockelee-Morvan et al. (2026)</a> |
|                  | $3.94 \times 10^{28}$                 | 1.30                        | 11/13/25          | 1.46          | SOHO/SWAN    | <a href="#">Tan et al. (2026)</a>             |
|                  | $2.45 \times 10^{29}$                 | 0.004                       | ...               | 1.47          | ...          | <a href="#">Combi et al. (2025)</a>           |
|                  | $2.09 \times 10^{29}$                 | 0.04                        | 11/14/25          | 1.48          | ...          | ...                                           |
|                  | $5.15 \times 10^{28}$                 | 1.65                        | 11/15/25          | 1.49          | ...          | <a href="#">Tan et al. (2026)</a>             |
|                  | $2.19 \times 10^{29}$                 | 0.004                       | ...               | 1.50          | ...          | <a href="#">Combi et al. (2025)</a>           |
|                  | $4.23 \times 10^{28}$                 | 1.37                        | 11/16/25          | 1.50          | ...          | <a href="#">Tan et al. (2026)</a>             |
|                  | $1.91 \times 10^{29}$                 | 0.004                       | ...               | 1.51          | ...          | <a href="#">Combi et al. (2025)</a>           |
|                  | $3.34 \times 10^{28}$                 | 1.07                        | 11/17/25          | 1.52          | ...          | <a href="#">Tan et al. (2026)</a>             |
|                  | $3.92 \times 10^{28}$                 | 1.27                        | 11/18/25          | 1.53          | ...          | ...                                           |
|                  | $2.51 \times 10^{29}$                 | 0.004                       | ...               | 1.55          | ...          | <a href="#">Combi et al. (2025)</a>           |
|                  | $4.70 \times 10^{28}$                 | 0.70                        | 11/19/25          | 1.56          | MAJIS/JUICE  | <a href="#">Bockelee-Morvan et al. (2026)</a> |
|                  | $3.90 \times 10^{29}$                 | 0.05                        | ...               | 1.56          | SOHO/SWAN    | <a href="#">Combi et al. (2025)</a>           |
|                  | $2.19 \times 10^{29}$                 | 0.004                       | 11/20/25          | 1.58          | ...          | ...                                           |
|                  | $2.55 \times 10^{28}$                 | 1.08                        | 11/25/25          | 1.67          | ...          | <a href="#">Tan et al. (2026)</a>             |
|                  | $4.10 \times 10^{28}$                 | 1.20                        | ...               | 1.68          | MAJIS/JUICE  | <a href="#">Bockelee-Morvan et al. (2026)</a> |
|                  | $1.83 \times 10^{28}$                 | 0.66                        | 11/26/25          | 1.69          | SOHO/SWAN    | <a href="#">Tan et al. (2026)</a>             |
|                  | $1.43 \times 10^{29}$                 | 0.004                       | ...               | 1.68          | ...          | <a href="#">Combi et al. (2025)</a>           |
|                  | $4.08 \times 10^{28}$                 | 0.07                        | 12/02/25          | 1.84          | ...          | ...                                           |
|                  | $2.46 \times 10^{28}$                 | 0.76                        | 12/03/25          | 1.86          | ...          | <a href="#">Tan et al. (2026)</a>             |
|                  | $7.60 \times 10^{27}$                 | 2.70                        | ...               | 1.86          | ...          | <a href="#">Combi et al. (2025)</a>           |
|                  | $1.55 \times 10^{28}$                 | 0.66                        | 12/04/25          | 1.89          | ...          | <a href="#">Tan et al. (2026)</a>             |
|                  | $2.43 \times 10^{28}$                 | 0.10                        | ...               | 1.89          | ...          | <a href="#">Combi et al. (2025)</a>           |
|                  | $1.88 \times 10^{28}$                 | 0.66                        | 12/05/25          | 1.91          | ...          | <a href="#">Tan et al. (2026)</a>             |
|                  | $1.14 \times 10^{28}$                 | 0.22                        | ...               | 1.91          | ...          | <a href="#">Combi et al. (2025)</a>           |
|                  | $1.30 \times 10^{28}$                 | 0.45                        | 12/06/25          | 1.94          | ...          | <a href="#">Tan et al. (2026)</a>             |
|                  | $3.68 \times 10^{28}$                 | 0.08                        | ...               | 1.94          | ...          | <a href="#">Combi et al. (2025)</a>           |
|                  | $1.26 \times 10^{28}$                 | 0.51                        | 12/07/25          | 1.96          | ...          | <a href="#">Tan et al. (2026)</a>             |
|                  | $2.04 \times 10^{28}$                 | 0.14                        | ...               | 1.96          | SOHO/SWAN    | <a href="#">Combi et al. (2025)</a>           |
|                  | $1.50 \times 10^{28}$                 | 0.54                        | 12/08/25          | 1.99          | ...          | <a href="#">Tan et al. (2026)</a>             |
|                  | $1.40 \times 10^{28}$                 | 0.28                        | 12/08/25–12/15/25 | 2.00–2.20     | SPHEREx      | <a href="#">Lisse et al. (2026)</a>           |
|                  | $1.00 \times 10^{28}$                 | 0.25                        | 12/09/25          | 1.99          | SWAN-SOHO    | <a href="#">Combi et al. (2025)</a>           |
|                  | $1.60 \times 10^{28}$                 | 0.13                        | 12/10/25          | 2.01          | ...          | ...                                           |
|                  | $1.13 \times 10^{28}$                 | 0.42                        | 12/15/25          | 2.18          | ...          | <a href="#">Tan et al. (2026)</a>             |
|                  | $1.13 \times 10^{28}$                 | 0.48                        | 12/16/25          | 2.22          | SWAN-SOHO    | <a href="#">Tan et al. (2026)</a>             |
|                  | $3.78 \times 10^{27}$                 | 0.03                        | 12/16/25          | 2.20          | JWST/MIRI    | <a href="#">Belyakov et al. (2026)</a>        |

|                 |                         |      |                   |             |              |                                                   |
|-----------------|-------------------------|------|-------------------|-------------|--------------|---------------------------------------------------|
|                 | $1.28 \times 10^{28}$   | 0.47 | 12/17/25          | 2.24        | SWAN-SOHO    | <a href="#">Tan et al. (2026)</a>                 |
|                 | $1.61 \times 10^{27}$   | 0.01 | 12/22/25          | 2.40        | JWST/NIRSpec | <a href="#">Cordiner et al. (2026)</a>            |
|                 | $8.70 \times 10^{26}$   | 0.05 | ...               | 2.40        | ...          | <a href="#">Roth et al. (2026a)</a>               |
|                 | $1.57 \times 10^{27}$   | 0.01 | ...               | 2.40        | ...          | <a href="#">Roth et al. (2026b)</a>               |
|                 | $9.82 \times 10^{26}$   | 0.13 | 12/23/25          | 2.40        | ...          | <a href="#">Roth et al. (2026a)</a>               |
|                 | $1.55 \times 10^{27}$   | 0.12 | ...               | 2.40        | ...          | <a href="#">Roth et al. (2026b)</a>               |
|                 | $1.69 \times 10^{27}$   | 0.05 | 12/23/25          | 2.40        | ...          | <a href="#">Cordiner et al. (2026)</a>            |
|                 | $1.05 \times 10^{27}$   | 0.02 | 12/27/25          | 2.54        | JWST/MIRI    | <a href="#">Belyakov et al. (2026)</a>            |
| CO <sub>2</sub> | $1.70 \times 10^{26}$   | 0.01 | 08/06/25          | 3.32        | JWST/NIRSpec | <a href="#">Cordiner et al. (2025)</a>            |
|                 | $1.60 \times 10^{27}$   | 0.10 | 08/01/25–08/15/25 | 3.20        | SPHEREx      | <a href="#">Lisse et al. (2025a)</a>              |
|                 | $9.40 \times 10^{26}$   | 0.30 | 08/08/25–08/12/25 | 3.10–3.30   | SPHEREx      | <a href="#">Lisse et al. (2025b)</a>              |
|                 | $8.20 \times 10^{27}$   | 0.80 | 11/02/25          | 1.36        | MAJIS/JUICE  | <a href="#">Bockelee-Morvan et al. (2026)</a>     |
|                 | $6.90 \times 10^{27}$   | 1.00 | 11/12/25          | 1.45        | ...          | ...                                               |
|                 | $5.90 \times 10^{27}$   | 1.40 | 11/19/25          | 1.56        | ...          | ...                                               |
|                 | $4.00 \times 10^{27}$   | 0.70 | 11/25/25          | 1.68        | ...          | ...                                               |
|                 | $3.00 \times 10^{27}$   | 0.45 | 12/08/25–12/15/25 | 2.00–2.20   | SPHEREx      | <a href="#">Lisse et al. (2026)</a>               |
|                 | $8.70 \times 10^{27}$   | 0.09 | 12/16/25          | 2.20        | JWST/MIRI    | <a href="#">Belyakov et al. (2026)</a>            |
|                 | $1.76 \times 10^{27}$   | 0.01 | 12/23/25          | 2.40        | JWST/NIRSpec | <a href="#">Cordiner et al. (2026)</a>            |
|                 | $1.59 \times 10^{27}$   | 0.01 | ...               | 2.40        | ...          | <a href="#">Roth et al. (2026b)</a>               |
|                 | $5.42 \times 10^{27}$   | 0.06 | 12/27/25          | 2.54        | JWST/MIRI    | <a href="#">Belyakov et al. (2026)</a>            |
| CO              | $< 1.10 \times 10^{27}$ | 0.00 | 07/18/25–07/21/25 | 3.94 – 3.84 | JCMT         | <a href="#">Hinkle et al. (2025)</a>              |
|                 | $3.70 \times 10^{26}$   | 0.20 | 08/06/25          | 3.32        | JWST/NIRSpec | <a href="#">Cordiner et al. (2025)</a>            |
|                 | $1.00 \times 10^{26}$   | 0.25 | 08/01/25–08/15/25 | 3.20        | SPHEREx      | <a href="#">Lisse et al. (2025a)</a>              |
|                 | $< 2.80 \times 10^{26}$ | 0.00 | 08/08/25–08/12/25 | 3.10–3.30   | ...          | <a href="#">Lisse et al. (2025b)</a>              |
|                 | $5.75 \times 10^{27}$   | 1.91 | 09/07/25–09/29/25 | 2.01        | 13.7 m MWT   | <a href="#">Li et al. (2026)</a>                  |
|                 | $6.80 \times 10^{27}$   | 0.11 | 11/01/25–11/03/25 | 1.36–1.37   | IRAM         | <a href="#">Biver et al. (2026)</a>               |
|                 | $7.60 \times 10^{27}$   | 1.50 | 12/08/25–12/15/25 | 2.00–2.20   | SPHEREx      | <a href="#">Lisse et al. (2026)</a>               |
|                 | $3.50 \times 10^{27}$   | 0.02 | 12/23/25          | 2.40        | JWST/NIRSpec | <a href="#">Roth et al. (2026a)</a>               |
|                 | $3.82 \times 10^{27}$   | 0.01 | ...               | 2.40        | ...          | <a href="#">Roth et al. (2026b)</a>               |
|                 | $3.94 \times 10^{27}$   | 0.01 | 12/23/25          | 2.40        | ...          | <a href="#">Cordiner et al. (2026)</a>            |
| CN              | $< 5.60 \times 10^{24}$ | 0.00 | 07/02/25          | 4.47        | TTT          | <a href="#">de la Fuente Marcos et al. (2025)</a> |
|                 | $< 5.60 \times 10^{23}$ | 0.00 | 07/04/25–07/05/25 | 4.41        | X-SHOOTER    | <a href="#">Alvarez-Candal et al. (2025)</a>      |
|                 | $< 9.80 \times 10^{23}$ | 0.00 | 07/12/25          | 4.14        | SNIFS        | <a href="#">Hoogendam et al. (2025b)</a>          |
|                 | $< 9.77 \times 10^{23}$ | 0.00 | 07/23/25          | 3.78        | X-SHOOTER    | <a href="#">Rahatgaonkar et al. (2025)</a>        |
|                 | $1.29 \times 10^{24}$   | 1.05 | 07/27/25          | 3.65        | ...          | ...                                               |
|                 | $1.38 \times 10^{24}$   | 1.05 | 07/30/25          | 3.55        | ...          | ...                                               |
|                 | $5.25 \times 10^{24}$   | 1.10 | 08/09/25          | 3.23        | ...          | ...                                               |
|                 | $5.80 \times 10^{24}$   | 1.90 | 08/10/25          | 3.16        | MDM          | <a href="#">Salazar Manzano et al. (2025)</a>     |
|                 | $1.70 \times 10^{24}$   | 1.99 | 08/12/25          | 3.14        | UVES         | <a href="#">Rahatgaonkar et al. (2025)</a>        |
|                 | $4.82 \times 10^{24}$   | 0.34 | ...               | 3.14        | MDM          | <a href="#">Salazar Manzano et al. (2025)</a>     |
|                 | $1.58 \times 10^{24}$   | 1.99 | ...               | 3.14        | UVES         | <a href="#">Hutsemékers et al. (2026b)</a>        |
|                 | $3.80 \times 10^{24}$   | 1.02 | 08/14/25          | 3.07        | X-SHOOTER    | <a href="#">Rahatgaonkar et al. (2025)</a>        |
|                 | $3.89 \times 10^{24}$   | 1.12 | 08/15/25          | 3.04        | UVES         | <a href="#">Rahatgaonkar et al. (2025)</a>        |
|                 | $3.89 \times 10^{24}$   | 1.12 | ...               | 3.04        | X-SHOOTER    | <a href="#">Hutsemékers et al. (2026b)</a>        |
|                 | $4.17 \times 10^{24}$   | 1.02 | 08/16/25          | 3.01        | ...          | <a href="#">Rahatgaonkar et al. (2025)</a>        |
|                 | $6.34 \times 10^{24}$   | 0.29 | 08/16/25          | 2.97        | MDM          | <a href="#">Salazar Manzano et al. (2025)</a>     |
|                 | $7.17 \times 10^{24}$   | 0.38 | 08/17/25          | 2.94        | ...          | ...                                               |
|                 | $8.40 \times 10^{23}$   | 3.20 | 08/18/25          | 2.94        | SNIFS        | <a href="#">Hoogendam et al. (2025b)</a>          |
|                 | $6.46 \times 10^{24}$   | 1.02 | 08/21/25          | 2.85        | X-SHOOTER    | <a href="#">Rahatgaonkar et al. (2025)</a>        |
|                 | $1.70 \times 10^{24}$   | 0.50 | 08/24/25          | 2.75        | KCWI         | <a href="#">Hoogendam et al. (2025a)</a>          |
|                 | $5.70 \times 10^{24}$   | 1.50 | 08/27/25          | 2.66        | LRIS         | <a href="#">Medler et al. (2026)</a>              |
|                 | $6.16 \times 10^{24}$   | 1.10 | 08/28/25          | 2.64        | UVES         | <a href="#">Hutsemékers et al. (2026b)</a>        |
|                 | $< 2.45 \times 10^{25}$ | 0.00 | ...               | 2.45        | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>                |
|                 | $9.12 \times 10^{24}$   | 1.07 | 09/3/25–09/4/25   | 2.44        | UVES         | <a href="#">Hutsemékers et al. (2026b)</a>        |
|                 | $< 3.55 \times 10^{25}$ | 0.00 | 09/03/25          | 2.42        | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>                |
|                 | $< 8.91 \times 10^{25}$ | 0.00 | 09/08/25          | 2.28        | ...          | <a href="#">Luco et al. (2026)</a>                |
|                 | $4.07 \times 10^{25}$   | 1.51 | 09/09/25          | 2.25        | ...          | <a href="#">Luco et al. (2026)</a>                |
|                 | $1.07 \times 10^{25}$   | 1.02 | 09/10/25          | 2.25        | UVES         | <a href="#">Hutsemékers et al. (2026b)</a>        |

|     |                         |      |                   |           |              |                                              |
|-----|-------------------------|------|-------------------|-----------|--------------|----------------------------------------------|
|     | $4.07 \times 10^{25}$   | 1.20 | ...               | 2.22      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $2.19 \times 10^{25}$   | 1.02 | 09/11/25          | 2.22      | UVES         | <a href="#">Hutsemékers et al. (2026b)</a>   |
|     | $3.24 \times 10^{25}$   | 1.38 | ...               | 2.19      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $2.88 \times 10^{25}$   | 1.02 | 09/12/25          | 2.19      | UVES         | <a href="#">Hutsemékers et al. (2026b)</a>   |
|     | $6.17 \times 10^{25}$   | 1.51 | ...               | 2.17      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $2.69 \times 10^{25}$   | 1.02 | 09/14/25          | 2.14      | UVES         | <a href="#">Hutsemékers et al. (2026b)</a>   |
|     | $1.17 \times 10^{26}$   | 1.17 | 09/15/25          | 2.08      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $1.41 \times 10^{26}$   | 1.41 | 09/16/25          | 2.06      | ...          | ...                                          |
|     | $6.31 \times 10^{25}$   | 1.17 | 09/17/25          | 3.03      | ...          | ...                                          |
|     | $6.03 \times 10^{25}$   | 1.15 | 09/18/25          | 2.00      | ...          | ...                                          |
|     | $3.63 \times 10^{25}$   | 1.17 | 09/19/25          | 1.98      | ...          | ...                                          |
|     | $4.07 \times 10^{25}$   | 1.05 | 09/20/25          | 1.97      | X-SHOOTER    | <a href="#">Hutsemékers et al. (2026b)</a>   |
|     | $7.76 \times 10^{25}$   | 1.15 | ...               | 1.95      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $5.01 \times 10^{25}$   | 1.05 | 09/22/25          | 1.92      | X-SHOOTER    | <a href="#">Hutsemékers et al. (2026b)</a>   |
|     | $4.90 \times 10^{25}$   | 1.20 | ...               | 1.90      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $5.25 \times 10^{25}$   | 1.05 | 09/23/25          | 1.90      | X-SHOOTER    | <a href="#">Hutsemékers et al. (2026b)</a>   |
|     | $7.94 \times 10^{25}$   | 1.20 | ...               | 1.87      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $5.37 \times 10^{25}$   | 1.05 | 09/24/25          | 1.87      | X-SHOOTER    | <a href="#">Hutsemékers et al. (2026b)</a>   |
|     | $1.55 \times 10^{25}$   | 1.15 | ...               | 1.85      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $5.37 \times 10^{25}$   | 1.05 | 09/25/25          | 1.85      | X-SHOOTER    | <a href="#">Hutsemékers et al. (2026b)</a>   |
|     | $1.60 \times 10^{26}$   | 0.50 | 11/16/25          | 1.50      | KCWI         | <a href="#">Hoogendam et al. (2026)</a>      |
|     | $8.91 \times 10^{25}$   | 1.29 | 11/25/25          | 1.67      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $6.17 \times 10^{25}$   | 1.38 | 11/26/25          | 1.70      | ...          | <a href="#">Luco et al. (2026)</a>           |
|     | $8.13 \times 10^{25}$   | 1.15 | 11/27/25          | 1.72      | ...          | <a href="#">Luco et al. (2026)</a>           |
|     | $3.20 \times 10^{25}$   | 1.20 | 11/30/25          | 1.79      | SNIFS        | <a href="#">Medler et al. (2026)</a>         |
|     | $2.18 \times 10^{25}$   | 1.05 | 12/02/25          | 1.85      | BFOSC        | <a href="#">Zhao et al. (2026)</a>           |
|     | $6.76 \times 10^{25}$   | 1.38 | 12/04/25          | 1.88      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $3.31 \times 10^{25}$   | 1.05 | ...               | 1.89      | YFOSC        | <a href="#">Zhao et al. (2026)</a>           |
|     | $8.91 \times 10^{25}$   | 1.29 | 12/06/25          | 1.93      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $3.09 \times 10^{25}$   | 1.05 | ...               | 1.95      | YFOSC        | <a href="#">Zhao et al. (2026)</a>           |
|     | $3.16 \times 10^{25}$   | 1.05 | 12/07/25          | 1.97      | ...          | ...                                          |
|     | $1.95 \times 10^{25}$   | 1.05 | 12/17/25          | 2.25      | ...          | ...                                          |
|     | $2.75 \times 10^{25}$   | 1.07 | 12/19/25          | 2.31      | ...          | ...                                          |
|     | $1.45 \times 10^{25}$   | 1.15 | 12/20/25          | 2.33      | BFOSC        | ...                                          |
|     | $< 3.39 \times 10^{25}$ | 0.00 | 12/21/25          | 2.35      | VLT/ESPRESSO | <a href="#">Luco et al. (2026)</a>           |
|     | $< 3.10 \times 10^{24}$ | 0.00 | 12/23/25          | 2.40      | JWST/NIRSpec | <a href="#">Cordiner et al. (2026)</a>       |
|     | $1.26 \times 10^{25}$   | 1.15 | 12/23/25          | 2.42      | BFOSC        | <a href="#">Zhao et al. (2026)</a>           |
|     | $8.91 \times 10^{24}$   | 1.15 | 12/26/25          | 2.51      | BFOSC        | <a href="#">Zhao et al. (2026)</a>           |
|     | $8.91 \times 10^{24}$   | 1.05 | 12/28/25          | 2.57      | YFOSC        | ...                                          |
|     | $6.03 \times 10^{24}$   | 1.07 | 1/11/26           | 3.00      | ...          | ...                                          |
|     | $4.37 \times 10^{24}$   | 1.07 | 1/20/26           | 3.29      | ...          | ...                                          |
| HCN | $< 1.70 \times 10^{24}$ | 0.00 | 07/16/25–07/17/25 | 4.01–3.97 | JCMT         | <a href="#">Hinkle et al. (2025)</a>         |
|     | $< 1.50 \times 10^{25}$ | 0.00 | 08/07/25          | 3.29      | JCMT         | <a href="#">Coulson et al. (2025)</a>        |
|     | $< 2.10 \times 10^{25}$ | 0.00 | 08/15/25          | 3.00      | ...          | ...                                          |
|     | $< 1.70 \times 10^{25}$ | 0.00 | 08/22/25          | 2.8       | ...          | ...                                          |
|     | $< 1.40 \times 10^{25}$ | 0.00 | 08/28/25          | 2.81      | ...          | ...                                          |
|     | $< 1.10 \times 10^{25}$ | 0.00 | 09/03/25          | 2.63      | ...          | ...                                          |
|     | $1.20 \times 10^{25}$   | 0.50 | 09/07/25          | 2.33      | ...          | ...                                          |
|     | $5.00 \times 10^{24}$   | 1.00 | 09/12/25          | 2.17      | ALMA /ACA    | <a href="#">Roth et al. (2025)</a>           |
|     | $4.00 \times 10^{25}$   | 1.70 | 09/14/25          | 2.14      | JCMT         | <a href="#">Coulson et al. (2025)</a>        |
|     | $1.00 \times 10^{25}$   | 0.10 | 09/15/25          | 2.11      | ALMA /ACA    | <a href="#">Roth et al. (2025)</a>           |
|     | $4.10 \times 10^{25}$   | 0.20 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>          |
|     | $6.75 \times 10^{24}$   | 0.54 | 12/22/25          | 2.37      | ALMA /ACA    | <a href="#">Cordiner et al. (2026)</a>       |
| OH  | $< 8.20 \times 10^{26}$ | 0.00 | 07/04/25–05/26/25 | 4.41      | X-SHOOTER    | <a href="#">Alvarez-Candal et al. (2025)</a> |
|     | $< 1.48 \times 10^{26}$ | 0.00 | 07/27/25          | 3.14      | UVES         | <a href="#">Rahatgaonkar et al. (2025)</a>   |
|     | $< 8.00 \times 10^{27}$ | 0.00 | 08/26/25–09/03/25 | 2.54      | TMRT         | <a href="#">Li et al. (2026)</a>             |
|     | $3.80 \times 10^{26}$   | 1.10 | 08/28/25          | 2.64      | UVES         | <a href="#">Hutsemékers et al. (2026b)</a>   |
|     | $6.16 \times 10^{26}$   | 1.05 | 09/03/25–09/04/25 | 2.44      | ...          | ...                                          |

|    |                         |      |                   |      |           |                            |
|----|-------------------------|------|-------------------|------|-----------|----------------------------|
|    | $1.32 \times 10^{28}$   | 0.47 | 09/09/25          | 2.27 | TMRT      | Li et al. (2026)           |
|    | $1.89 \times 10^{28}$   | 0.37 | 09/18/25–09/23/25 | 1.96 | ...       | ...                        |
|    | $1.74 \times 10^{27}$   | 1.35 | 09/10/25          | 2.25 | UVES      | Hutsemékers et al. (2026b) |
|    | $2.14 \times 10^{27}$   | 1.26 | 09/12/25          | 2.19 | ...       | ...                        |
| Ni | $1.62 \times 10^{22}$   | 1.35 | 07/23/25          | 3.78 | X-SHOOTER | Rahatgaonkar et al. (2025) |
|    | $1.29 \times 10^{22}$   | 1.62 | ...               | 3.78 | ...       | Hutsemékers et al. (2026b) |
|    | $2.75 \times 10^{22}$   | 1.44 | 07/27/25          | 3.65 | ...       | Rahatgaonkar et al. (2025) |
|    | $2.24 \times 10^{22}$   | 1.62 | ...               | 3.65 | ...       | Hutsemékers et al. (2026b) |
|    | $2.69 \times 10^{22}$   | 1.14 | 07/30/25          | 3.55 | ...       | ...                        |
|    | $3.71 \times 10^{22}$   | 1.32 | ...               | 3.55 | ...       | Rahatgaonkar et al. (2025) |
|    | $1.32 \times 10^{23}$   | 1.12 | 08/09/25          | 3.23 | ...       | ...                        |
|    | $6.76 \times 10^{22}$   | 1.05 | ...               | 3.23 | ...       | Hutsemékers et al. (2026b) |
|    | $6.61 \times 10^{22}$   | 1.15 | 08/12/25          | 3.14 | UVES      | ...                        |
|    | $6.60 \times 10^{22}$   | 1.15 | ...               | 3.14 | ...       | Rahatgaonkar et al. (2025) |
|    | $1.23 \times 10^{23}$   | 1.17 | 08/14/25          | 3.07 | X-SHOOTER | ...                        |
|    | $6.76 \times 10^{22}$   | 1.17 | ...               | 3.07 | ...       | Hutsemékers et al. (2026b) |
|    | $5.49 \times 10^{22}$   | 1.20 | 08/15/25          | 3.04 | UVES      | ...                        |
|    | $5.49 \times 10^{24}$   | 1.20 | ...               | 3.04 | ...       | Rahatgaonkar et al. (2025) |
|    | $1.41 \times 10^{24}$   | 1.20 | 08/16/25          | 3.01 | X-SHOOTER | ...                        |
|    | $8.13 \times 10^{22}$   | 1.15 | ...               | 3.01 | ...       | Hutsemékers et al. (2026b) |
|    | $1.99 \times 10^{22}$   | 1.17 | 08/21/25          | 2.85 | ...       | Rahatgaonkar et al. (2025) |
|    | $1.20 \times 10^{22}$   | 1.15 | ...               | 2.85 | X-SHOOTER | Hutsemékers et al. (2026b) |
|    | $1.86 \times 10^{23}$   | 1.12 | 08/28/25          | 2.64 | UVES      | ...                        |
|    | $3.23 \times 10^{23}$   | 1.09 | 09/03/25–09/04/25 | 2.44 | ...       | ...                        |
|    | $5.62 \times 10^{23}$   | 1.07 | 09/10/25          | 2.25 | ...       | ...                        |
|    | $6.30 \times 10^{23}$   | 1.07 | 09/12/25          | 2.19 | ...       | ...                        |
|    | $1.12 \times 10^{24}$   | 1.12 | 09/20/25          | 1.97 | X-SHOOTER | ...                        |
|    | $1.78 \times 10^{24}$   | 1.10 | 09/22/25          | 1.92 | ...       | ...                        |
|    | $1.99 \times 10^{24}$   | 1.10 | 09/23/25          | 1.90 | ...       | ...                        |
|    | $1.95 \times 10^{24}$   | 1.01 | 09/24/25          | 1.87 | ...       | ...                        |
|    | $2.04 \times 10^{24}$   | 1.01 | 09/25/25          | 1.85 | ...       | ...                        |
|    | $7.47 \times 10^{24}$   | 0.27 | 11/16/25          | 1.50 | KCWI      | Hoogendam et al. (2026)    |
|    | $9.77 \times 10^{24}$   | 1.05 | 12/02/25          | 1.85 | BFOSC     | Zhao et al. (2026)         |
|    | $8.71 \times 10^{24}$   | 1.23 | 12/04/25          | 1.89 | YFOSC     | ...                        |
|    | $7.76 \times 10^{24}$   | 1.23 | 12/06/25          | 1.95 | ...       | ...                        |
|    | $9.12 \times 10^{23}$   | 1.07 | ...               | 1.93 | UVES      | Hutsemékers et al. (2026a) |
|    | $6.76 \times 10^{24}$   | 1.23 | 12/07/25          | 1.97 | YFOSC     | Zhao et al. (2026)         |
|    | $9.33 \times 10^{23}$   | 1.10 | 12/10/25          | 2.04 | UVES      | Hutsemékers et al. (2026a) |
|    | $7.78 \times 10^{23}$   | 1.07 | 12/15/25          | 2.18 | ...       | ...                        |
|    | $7.76 \times 10^{24}$   | 1.14 | 12/16/25          | 2.20 | JWST/MIRI | Belyakov et al. (2026)     |
|    | $5.01 \times 10^{24}$   | 1.23 | 12/17/25          | 2.25 | YFOSC     | Zhao et al. (2026)         |
|    | $3.63 \times 10^{24}$   | 1.23 | 12/19/25          | 2.31 | ...       | ...                        |
|    | $4.7 \times 10^{24}$    | 1.05 | 12/20/25          | 2.33 | BFOSC     | ...                        |
|    | $6.16 \times 10^{23}$   | 1.10 | 12/21/25          | 2.35 | UVES      | Hutsemékers et al. (2026a) |
|    | $3.98 \times 10^{24}$   | 1.05 | 12/23/25          | 2.42 | BFOSC     | Zhao et al. (2026)         |
|    | $3.39 \times 10^{24}$   | 1.05 | 12/26/25          | 2.51 | ...       | ...                        |
|    | $4.57 \times 10^{23}$   | 1.10 | ...               | 2.50 | UVES      | Hutsemékers et al. (2026a) |
|    | $2.88 \times 10^{24}$   | 1.23 | 12/28/25          | 2.57 | YFOSC     | Zhao et al. (2026)         |
|    | $1.82 \times 10^{23}$   | 1.12 | 01/11/26          | 2.99 | UVES      | Hutsemékers et al. (2026a) |
|    | $1.95 \times 10^{24}$   | 1.23 | ...               | 3.00 | YFOSC     | Zhao et al. (2026)         |
|    | $1.14 \times 10^{23}$   | 1.15 | 01/19/26          | 3.24 | UVES      | Hutsemékers et al. (2026a) |
|    | $1.51 \times 10^{24}$   | 1.23 | 01/20/26          | 3.29 | YFOSC     | Zhao et al. (2026)         |
|    | $7.59 \times 10^{22}$   | 1.17 | 01/27/26          | 3.50 | UVES      | Hutsemékers et al. (2026a) |
|    | $2.75 \times 10^{22}$   | 1.26 | 02/07/26          | 3.85 | ...       | ...                        |
| Fe | $< 6.76 \times 10^{21}$ | 0.00 | 08/12/25          | 3.14 | UVES      | Hutsemékers et al. (2026b) |
|    | $< 1.38 \times 10^{22}$ | 0.00 | 08/15/25          | 3.04 | ...       | ...                        |
|    | $1.00 \times 10^{22}$   | 1.20 | 08/28/25          | 2.64 | ...       | ...                        |
|    | $4.00 \times 10^{22}$   | 1.10 | 09/03/25–09/04/25 | 2.44 | ...       | ...                        |

|                               |                         |      |                   |           |              |                                               |
|-------------------------------|-------------------------|------|-------------------|-----------|--------------|-----------------------------------------------|
|                               | $1.15 \times 10^{23}$   | 1.07 | 09/10/25          | 2.25      | ...          | ...                                           |
|                               | $1.58 \times 10^{23}$   | 1.05 | 09/12/15          | 2.19      | ...          | ...                                           |
|                               | $4.78 \times 10^{23}$   | 1.07 | 09/20/25          | 1.97      | X-SHOOTER    | ...                                           |
|                               | $8.51 \times 10^{23}$   | 1.07 | 09/22/25          | 1.92      | ...          | ...                                           |
|                               | $1.00 \times 10^{24}$   | 1.05 | 09/23/25          | 1.90      | ...          | ...                                           |
|                               | $1.15 \times 10^{24}$   | 1.07 | 09/24/25          | 1.87      | ...          | ...                                           |
|                               | $1.17 \times 10^{24}$   | 1.07 | 09/25/25          | 1.85      | ...          | ...                                           |
|                               | $6.30 \times 10^{24}$   | 0.10 | 11/16/25          | 1.50      | KCWI         | <a href="#">Hoogendam et al. (2026)</a>       |
|                               | $2.04 \times 10^{24}$   | 1.05 | 12/02/25          | 1.85      | BFOSC        | <a href="#">Zhao et al. (2026)</a>            |
|                               | $1.99 \times 10^{24}$   | 1.23 | 12/04/25          | 1.89      | YFOSC        | <a href="#">Zhao et al. (2026)</a>            |
|                               | $2.09 \times 10^{24}$   | 1.10 | 12/04/25          | 1.88      | UVES         | <a href="#">Hutsemékers et al. (2026a)</a>    |
|                               | $8.32 \times 10^{23}$   | 1.02 | 12/06/25          | 1.93      | ...          | ...                                           |
|                               | $1.74 \times 10^{24}$   | 1.23 | 12/06/25          | 1.95      | YFOSC        | <a href="#">Zhao et al. (2026)</a>            |
|                               | $1.48 \times 10^{24}$   | 1.23 | 12/07/25          | 1.97      | ...          | ...                                           |
|                               | $8.51 \times 10^{23}$   | 1.02 | 12/10/25          | 2.04      | UVES         | <a href="#">Hutsemékers et al. (2026a)</a>    |
|                               | $6.03 \times 10^{23}$   | 1.02 | 12/15/25          | 2.18      | ...          | ...                                           |
|                               | $5.62 \times 10^{24}$   | 1.23 | 12/17/25          | 2.25      | YFOSC        | <a href="#">Zhao et al. (2026)</a>            |
|                               | $4.07 \times 10^{24}$   | 1.23 | 12/19/25          | 2.31      | ...          | ...                                           |
|                               | $4.17 \times 10^{24}$   | 1.07 | 12/20/25          | 2.33      | BFOSC        | ...                                           |
|                               | $3.16 \times 10^{23}$   | 1.02 | 12/21/25          | 2.35      | UVES         | <a href="#">Hutsemékers et al. (2026a)</a>    |
|                               | $2.45 \times 10^{23}$   | 1.07 | 12/23/25          | 2.42      | BFOSC        | <a href="#">Zhao et al. (2026)</a>            |
|                               | $1.86 \times 10^{23}$   | 1.02 | 12/26/25          | 2.50      | UVES         | <a href="#">Hutsemékers et al. (2026a)</a>    |
|                               | $1.74 \times 10^{23}$   | 1.23 | 12/28/25          | 2.57      | YFOSC        | ...                                           |
|                               | $4.27 \times 10^{22}$   | 1.10 | 01/11/26          | 2.99      | UVES         | <a href="#">Hutsemékers et al. (2026a)</a>    |
|                               | $2.29 \times 10^{22}$   | 1.17 | 01/19/26          | 3.24      | ...          | ...                                           |
|                               | $1.38 \times 10^{22}$   | 1.50 | 01/27/26          | 3.50      | ...          | ...                                           |
| C <sub>2</sub>                | $2.47 \times 10^{24}$   | 0.82 | 08/10/25          | 3.16      | MDM          | <a href="#">Salazar Manzano et al. (2025)</a> |
|                               | $2.84 \times 10^{24}$   | 0.95 | 08/12/25          | 3.14      | ...          | ...                                           |
|                               | $1.43 \times 10^{24}$   | 0.48 | 08/16/25          | 2.97      | ...          | ...                                           |
|                               | $1.43 \times 10^{24}$   | 0.48 | 08/17/25          | 2.94      | ...          | ...                                           |
|                               | $3.98 \times 10^{24}$   | 1.58 | 08/28/25          | 2.64      | UVES         | <a href="#">Hutsemékers et al. (2026b)</a>    |
|                               | $3.98 \times 10^{24}$   | 1.58 | 09/03/25–09/04/25 | 2.44      | ...          | ...                                           |
|                               | $3.71 \times 10^{24}$   | 1.05 | 09/10/25          | 2.25      | ...          | ...                                           |
|                               | $4.78 \times 10^{24}$   | 1.12 | 09/12/25          | 2.19      | ...          | ...                                           |
|                               | $1.74 \times 10^{25}$   | 1.12 | 09/22/25          | 1.92      | X-SHOOTER    | ...                                           |
|                               | $2.09 \times 10^{25}$   | 1.12 | 09/23/25          | 1.90      | ...          | ...                                           |
|                               | $2.57 \times 10^{25}$   | 1.12 | 09/24/25          | 1.87      | ...          | ...                                           |
|                               | $2.57 \times 10^{25}$   | 1.12 | 09/25/25          | 1.85      | ...          | ...                                           |
|                               | $8.80 \times 10^{25}$   | 0.80 | 11/16/25          | 1.50      | KCWI         | <a href="#">Hoogendam et al. (2026)</a>       |
|                               | $6.46 \times 10^{24}$   | 1.05 | 12/02/25          | 1.85      | BFOSC        | <a href="#">Zhao et al. (2026)</a>            |
|                               | $6.31 \times 10^{24}$   | 1.23 | 12/04/25          | 1.89      | YFOSC        | ...                                           |
|                               | $5.75 \times 10^{24}$   | 1.23 | 12/06/25          | 1.95      | ...          | ...                                           |
|                               | $4.79 \times 10^{24}$   | 1.23 | 12/07/25          | 1.97      | ...          | ...                                           |
|                               | $2.00 \times 10^{24}$   | 1.23 | 12/17/25          | 2.25      | ...          | ...                                           |
|                               | $1.48 \times 10^{24}$   | 1.23 | 12/19/25          | 2.31      | ...          | ...                                           |
|                               | $1.48 \times 10^{24}$   | 1.07 | 12/20/25          | 2.33      | BFOSC        | ...                                           |
|                               | $8.32 \times 10^{23}$   | 1.07 | 12/23/25          | 2.42      | ...          | ...                                           |
| CH <sub>3</sub> OH            | $5.00 \times 10^{26}$   | 0.60 | 08/28/25          | 2.64      | ALMA /ACA    | <a href="#">Roth et al. (2025)</a>            |
|                               | $9.60 \times 10^{26}$   | 0.70 | 09/18/25          | 2.03      | ALMA /ACA    | <a href="#">Roth et al. (2025)</a>            |
|                               | $1.21 \times 10^{27}$   | 0.05 | 09/22/25          | 1.92      | ALMA /ACA    | <a href="#">Roth et al. (2025)</a>            |
|                               | $2.29 \times 10^{27}$   | 0.40 | 10/01/25          | 1.71      | ALMA /ACA    | <a href="#">Roth et al. (2025)</a>            |
|                               | $3.38 \times 10^{27}$   | 0.05 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
|                               | $4.10 \times 10^{27}$   | 0.10 | 11/04/25          | 1.37      | ALMA /ACA    | <a href="#">Salazar Manzano et al. (2026)</a> |
|                               | $1.37 \times 10^{26}$   | 0.04 | 12/23/25          | 2.40      | JWST/NIRSpec | <a href="#">Roth et al. (2026a)</a>           |
| <sup>13</sup> CO <sub>2</sub> | $2.65 \times 10^{26}$   | 0.03 | ...               | 2.40      | ...          | <a href="#">Roth et al. (2026b)</a>           |
|                               | $1.30 \times 10^{25}$   | 0.25 | 08/01/25–08/15/25 | 3.20      | SPHEREx      | <a href="#">Lisse et al. (2025a)</a>          |
|                               | $< 9.40 \times 10^{24}$ | 0.00 | 12/23/25          | 2.40      | JWST/NIRSpec | <a href="#">Cordiner et al. (2026)</a>        |
| <sup>13</sup> CO              | $< 2.49 \times 10^{25}$ | 0.00 | 12/23/25          | 2.40      | JWST/NIRSpec | <a href="#">Cordiner et al. (2026)</a>        |

|                                   |                         |      |                   |           |              |                                               |
|-----------------------------------|-------------------------|------|-------------------|-----------|--------------|-----------------------------------------------|
| OCS                               | $1.70 \times 10^{24}$   | 0.90 | 08/06/25          | 3.32      | JWST         | <a href="#">Cordiner et al. (2025)</a>        |
|                                   | $< 5.00 \times 10^{26}$ | 0.00 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
|                                   | $4.70 \times 10^{24}$   | 0.50 | 12/23/25          | 2.40      | JWST/NIRSpec | <a href="#">Cordiner et al. (2026)</a>        |
| C <sub>3</sub>                    | $1.20 \times 10^{24}$   | 0.29 | 08/10/25          | 3.16      | MDM          | <a href="#">Salazar Manzano et al. (2025)</a> |
|                                   | $8.90 \times 10^{23}$   | 3.00 | 08/12/25          | 3.14      | ...          | ...                                           |
|                                   | $8.90 \times 10^{23}$   | 3.00 | 08/16/25          | 2.97      | ...          | ...                                           |
|                                   | $9.40 \times 10^{23}$   | 3.10 | 08/17/25          | 2.94      | ...          | ...                                           |
|                                   | $5.00 \times 10^{24}$   | 0.40 | 11/16/25          | 1.50      | KCWI         | <a href="#">Hoogendam et al. (2026)</a>       |
|                                   | $1.86 \times 10^{24}$   | 1.05 | 12/02/25          | 1.85      | BFOSC        | <a href="#">Zhao et al. (2026)</a>            |
|                                   | $2.29 \times 10^{24}$   | 1.05 | 12/04/25          | 1.89      | YFOSC        | ...                                           |
|                                   | $2.29 \times 10^{24}$   | 1.05 | 12/06/25          | 1.95      | ...          | ...                                           |
|                                   | $2.04 \times 10^{24}$   | 1.05 | 12/07/25          | 1.97      | ...          | ...                                           |
|                                   | $1.07 \times 10^{24}$   | 1.05 | 12/17/25          | 2.25      | ...          | ...                                           |
|                                   | $1.62 \times 10^{24}$   | 1.07 | 12/19/25          | 2.31      | ...          | ...                                           |
|                                   | $1.66 \times 10^{24}$   | 1.07 | 12/20/25          | 2.33      | BFOSC        | ...                                           |
|                                   | $1.95 \times 10^{23}$   | 1.07 | 12/23/25          | 2.42      | ...          | ...                                           |
| misc. organics                    | $2.00 \times 10^{27}$   | 0.50 | 12/08/25–12/15/25 | 2.0–2.20  | SPHEREx      | <a href="#">Lisse et al. (2026)</a>           |
| CH <sub>4</sub>                   | $4.20 \times 10^{26}$   | 0.20 | 12/16/25          | 2.20      | JWST/MIRI    | <a href="#">Belyakov et al. (2026)</a>        |
|                                   | $1.90 \times 10^{26}$   | 0.02 | 12/23/25          | 2.40      | JWST/NIRSpec | <a href="#">Roth et al. (2026a)</a>           |
|                                   | $2.06 \times 10^{26}$   | 0.01 | ...               | 2.40      | ...          | <a href="#">Roth et al. (2026b)</a>           |
|                                   | $2.30 \times 10^{26}$   | 0.10 | 12/27/25          | 2.54      | JWST/MIRI    | <a href="#">Belyakov et al. (2026)</a>        |
| CH                                | $3.24 \times 10^{25}$   | 1.07 | 12/02/25          | 1.85      | BFOSC        | <a href="#">Zhao et al. (2026)</a>            |
|                                   | $4.36 \times 10^{25}$   | 1.07 | 12/04/25          | 1.89      | YFOSC        | ...                                           |
|                                   | $4.07 \times 10^{25}$   | 1.07 | 12/06/25          | 1.95      | ...          | ...                                           |
|                                   | $5.25 \times 10^{25}$   | 1.07 | 12/07/25          | 1.97      | ...          | ...                                           |
|                                   | $2.34 \times 10^{25}$   | 1.10 | 12/17/25          | 2.25      | ...          | ...                                           |
|                                   | $2.69 \times 10^{25}$   | 1.20 | 12/19/25          | 2.31      | ...          | ...                                           |
| HDO                               | $< 1.21 \times 10^{27}$ | 0.00 | 11/01/25–11/3/25  | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
|                                   | $3.20 \times 10^{25}$   | 0.19 | 12/23/25          | 2.40      | JWST/NIRSpec | <a href="#">Cordiner et al. (2026)</a>        |
| CH <sub>3</sub> CN                | $1.50 \times 10^{25}$   | 0.40 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
| H <sub>2</sub> CO                 | $3.50 \times 10^{26}$   | 0.40 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
|                                   | $3.90 \times 10^{24}$   | 0.90 | 12/23/15          | 2.40      | JWST/NIRSpec | <a href="#">Roth et al. (2026a)</a>           |
| H <sub>2</sub> S                  | $< 3.90 \times 10^{26}$ | 0.00 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
| CS                                | $1.70 \times 10^{25}$   | 0.40 | 11/01/25–11/30/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
| HC <sub>3</sub> N                 | $< 2.10 \times 10^{25}$ | 0.00 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
| CH <sub>3</sub> CHO               | $1.40 \times 10^{26}$   | 0.70 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
| (CH <sub>3</sub> OH) <sub>2</sub> | $3.30 \times 10^{26}$   | 1.30 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
| SO                                | $< 2.70 \times 10^{26}$ | 0.00 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
| HCNO                              | $2.50 \times 10^{26}$   | 1.10 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
| HCOOH                             | $< 1.09 \times 10^{27}$ | 0.00 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
| NH <sub>2</sub> CHO               | $< 4.80 \times 10^{25}$ | 0.00 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
| CH <sub>3</sub> OD                | $< 1.79 \times 10^{26}$ | 0.00 | 11/01/25–11/03/25 | 1.36–1.37 | IRAM         | <a href="#">Biver et al. (2026)</a>           |
| CH <sub>3</sub> D                 | $2.50 \times 10^{25}$   | 0.30 | 12/23/25          | 2.40      | JWST/NIRSpec | <a href="#">Roth et al. (2026a)</a>           |
| C <sub>2</sub> H <sub>6</sub>     | $2.00 \times 10^{25}$   | 0.20 | 12/23/25          | 2.40      | JWST/NIRSpec | <a href="#">Roth et al. (2026a)</a>           |

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